- You have to deal with dependencies (e.g., wait for the instance to be running before connecting to it).
- There is no easy way to update the instances that are running with new settings (e.g., change the instance type).

It's time to enter a new world by leaving the imperative world and entering the declarative world.

4.4 Infrastructure as Code

Infrastructure as Code is the idea of using a high-level programming language to control infrastructures. Infrastructure can be any AWS resource, like a network topology, a load balancer, a DNS entry, and so on. In software development, tools like automated tests, code repositories, and build servers increase the quality of software engineering. If your infrastructure is defined as code, then you can apply these types of software development tools to your infrastructure and improve its quality.

WARNING Don't mix up the terms Infrastructure as Code and *Infrastructure as a Service* (IaaS)! IaaS means renting virtual machines, storage, and network with a pay-per-use pricing model.

4.4.1 Inventing an infrastructure language: JIML

For the purposes of learning the concepts behind Infrastructure as Code, let's invent a new language to describe infrastructure: JSON Infrastructure Markup Language (JIML). Figure 4.12 shows the infrastructure that will be created in the end.